

Yuvraj Chahal

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EDUCATION

B.S. in Computer Science + Economics - GPA: 3.8/4.0

Expected Graduation, May 2026

University of Illinois at Urbana-Champaign (UIUC)

- **Related Coursework:** Data Structures & Algorithms, Applied Machine Learning, Advanced Algorithms, Computer Systems, Distributed Systems, Languages & Compilers, Databases, Discrete Math, Linear Algebra, Computer Vision

EXPERIENCE

Software Development Engineer Intern

Amazon

May 2025 – Present

- Incoming Summer 2025

Software Engineering Intern

Zolon Tech, Inc.

June 2024 – Aug 2024

- Automated onboarding for the Administration for Children & Families, increasing document digitization efficiency by **~30%** by developing a **machine learning solution** utilizing **optical character recognition** in an agile environment.
- Enabled seamless data integration for **1,000+** ACF employees by engineering **Python AWS Lambda functions** using Amazon **Textract**, stored **JSON** data in **Amazon S3**, and supported **REST API** integration with existing processes.
- Achieved **85% accuracy** in detecting incorrect previous case closeouts by implementing a **facial recognition system** using a **Siamese neural network** and **FaceNet**, through **data augmentation and feature extraction** techniques.

Computer Science Course Assistant

University of Illinois at Urbana-Champaign (UIUC)

Jan 2023 – Dec 2023

- Guided 100+ students in **debugging, testing**, & understanding complex **object-oriented concepts** during office hours
- Personalized support to each student, enhancing comprehension & performance in homework, projects, and quizzes.

PROJECTS

Neural Network from Scratch

- Designed a **neural network** from scratch using **PyTorch** to play Pac-Man
- Developed a **deep convolutional Q-learning agent** with multiple **convolutional layers, batch normalization, and fully connected layers**, leveraging experience replay and target networks for stable and efficient learning.
- Fine-tuned hyperparameters such as learning rate, discount factor, and epsilon decay, employing **Adam optimizer** and **mean squared error loss** for optimal Q-value estimation.

HackIllinois 2024 Hardware Hack

- Won **2nd place** out of **800+ competitors** engineering a robot using **Computer Vision** and Raspberry Pi.
- Designed and implemented **autonomous navigation** logic in Python, incorporating **state machine principles** for dynamic behavior adjustment in real-world conditions.
- Utilized **OpenCV** for **real-time image processing**, enabling the robot to follow paths and detect/avoid obstacles using **color filtering and contour detection** techniques.

TwitterNLP

- Developed a Python-based **sentiment analysis tool** to scrape tweets from certain users, search terms, or hashtags.
- Utilized the advanced RoBERTa model, an evolution of Google's BERT, for **natural language processing**, accurately categorizing tweets into positive, neutral, and negative sentiments.
- Implemented **data visualization** using **Plotly**, showcasing sentiment trends over time and enabling detailed analysis through user-interactive features.

Illini Rideshare

- Created a web application using **React.js, Node.js**, and **HTML/CSS/JS** for university ridesharing.
- Integrated **Microsoft's Authentication Library** for secure access, restricting sign-in to University of Illinois students.

SKILLS

Programming: Java, Python, C++, C, SQL, MySQL, MongoDB, JavaScript, HTML/CSS, Node.js, React.js, R

Tools: Git, Scikit-learn (sklearn), PyTorch, OpenCV, Pandas, NumPy, Matplotlib, Amazon Web Services (AWS), Docker, Flask